6.2

1 (1) $2\sin^{-1}\frac{\sqrt{3}}{2}$

$= 2 \cdot \frac{\pi}{3}$

$= \frac{2}{3}\pi$

2 (1)

$$\lim_{x \to \infty} \tan^{-1} x = \frac{\pi}{2}$$

6 (1) $y = \tan^{-1}\frac{2x}{1-x^2}$

$$y' = \frac{\frac{2(1-x^2) - 2x \cdot (-2x)}{(1-x^2)^2}}{1 + \left(\frac{2x}{1-x^2}\right)^2}$$

$$= \frac{\frac{2x^2+2}{(1-x^2)^2}}{1 + \left(\frac{2x}{1-x^2}\right)^2} = \frac{\frac{2(x^2+1)}{(1-x^2)^2}}{\frac{(1-x^2)^2 + 4x^2}{(1-x^2)^2}}$$

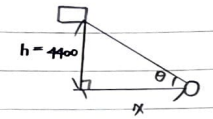
$$= \frac{2(x^2+1)}{(x^2+1)^2}$$

$$= \frac{2}{x^2+1}$$

(3) $f(x) = \cos^{-1}\frac{2}{x}$

$$f'(x) = +\frac{\frac{2}{x^2}}{\sqrt{1 - \frac{4}{x^2}}} = \frac{2}{x^2\sqrt{1 - \frac{4}{x^2}}}$$

10 $\frac{d\theta}{dt} = -0.05$



$\tan\theta = \frac{4400}{x}$

$\sec^2\theta \cdot \frac{d\theta}{dt} = -\frac{4400}{x^2} \cdot \frac{dx}{dt}$

$\left(\theta = \frac{\pi}{4} \quad \frac{d\theta}{dt} = -0.05 \quad x = 4400\right)$

$\Downarrow$

$\frac{1}{\cos^2\frac{\pi}{4}} \cdot (-0.05) = -\frac{1}{4400} \cdot \frac{dx}{dt}$

$\frac{dx}{dt} = (-4400) \cdot 2 \cdot (-0.05)$

$= 440 \text{ ft/s}$

6.3

1 (2) $y = (\cos x)^{\cos x}$

$\ln y = \cos x \cdot \ln(\cos x)$

$\frac{1}{y} \cdot y' = -\sin x \cdot \ln(\cos x) - \sin x$

$y' = -\sin x\left(\ln(\cos x) + 1\right) \cdot (\cos x)^{\cos x}$

(5) $(2x+3)^4 \, y = x^{\frac{1}{3}}\sqrt{x^2+1}$

$\ln(2x+3)^4 + \ln y = \ln x^{\frac{1}{3}} + \ln\sqrt{x^2+1}$

$\frac{4(2x+3)^3 \cdot 2}{(2x+3)^4} + \frac{1}{y} \cdot y' = \frac{1}{3} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{2x}{x^2+1}$

$\frac{8}{2x+3} + \frac{1}{y} \cdot y' = \frac{1}{3x} + \frac{x}{x^2+1}$

$\frac{1}{y} \cdot y' = \frac{1}{3x} + \frac{x}{x^2+1} - \frac{8}{2x+3}$

$\therefore y' = \frac{x^{\frac{1}{3}}\sqrt{x^2+1}}{(2x+3)^4}\left(\frac{1}{3x} + \frac{x}{x^2+1} - \frac{8}{2x+3}\right)$

8 (3) $y = x^{e^x}$

$\ln y = e^x \cdot \ln x$

$\frac{1}{y} \cdot y' = e^x \ln x + \frac{e^x}{x}$

$\therefore \frac{dy}{dx} = x^{e^x}\left(e^x \ln x + \frac{e^x}{x}\right)$

10 (4) $e^{-\cos^{-1}x}$

$e^{-\cos^{-1}x} \cdot \frac{1}{\sqrt{1-x^2}}$